

WIARTON, ON, Canada

WMO: 716330

Lat: 44.748N

Lon: 81.111W

Elev: 729

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	2.0	-11.0	3.1	-1.8	-5.4	4.2	3.0	27.5	29.0	24.4	29.6	6.4	140	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	83.4	71.4	80.5	69.6	77.9	67.9	73.1	80.6	71.4	78.1	69.7	75.6	10.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.7	116.0	77.2	69.0	109.6	75.4	67.5	103.7	73.8	37.2	81.0	35.7	78.1	34.2	75.5	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	20.2	18.1	-11.2	88.5	8.5	2.4	-17.4	90.2	-22.3	91.6	-27.1	93.0	-33.3	94.8
			-11.7	76.7	8.4	1.5	-17.8	77.8	-22.7	78.7	-27.4	79.5	-33.6	80.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	21.1	21.3	29.4	41.1	52.5	61.5	66.5	65.7	59.5	48.8	37.8	28.3	(6)
	DBStd	18.26	11.11	10.58	10.55	8.54	7.92	6.76	5.46	5.29	7.19	8.08	8.42	8.74	(7)
	HDD50	3876	897	802	645	291	62	2	0	0	7	122	374	674	(8)
	HDD65	7711	1362	1222	1105	718	398	147	48	56	195	505	816	1139	(9)
	CDD50	1898	0	0	6	23	139	347	511	485	292	86	8	1	(10)
	CDD65	257	0	0	0	0	9	43	94	77	30	4	0	0	(11)
	CDH74	1670	0	0	1	8	99	312	610	443	184	13	0	0	(12)
	CDH80	284	0	0	0	0	14	62	104	77	27	0	0	0	(13)
	WSAvg	8.2	10.0	9.5	8.7	8.9	7.3	6.0	6.3	6.5	7.2	8.9	9.7	9.8	(14)
	PrecAvg	41.3	4.1	2.8	3.2	2.7	2.8	2.9	2.9	3.6	4.3	3.5	4.3	5.0	(15)
(16) Precipitation	PrecMax	50.7	8.1	6.1	6.6	4.6	7.0	5.1	8.1	7.8	7.9	6.5	9.7	8.7	(16)
	PrecMin	27.1	2.1	0.9	0.5	0.6	0.8	0.3	0.0	1.0	1.3	1.3	1.4	2.6	(17)
	PrecStd	5.9	1.5	1.3	1.3	1.0	1.4	1.2	2.0	1.9	1.5	1.3	1.7	1.6	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	49.9	65.9	73.9	81.7	85.5	86.2	86.6	83.7	75.8	63.5	54.3	(19)
		MCWB	48.0	47.3	57.2	60.6	68.5	72.8	73.2	72.2	71.3	65.8	56.3	52.4	(20)
	2%	DB	43.5	42.7	55.4	66.6	76.8	81.5	82.9	81.9	79.1	70.5	58.4	47.0	(21)
		MCWB	41.6	39.7	49.9	55.5	64.7	69.8	71.8	71.1	68.6	62.4	53.3	44.4	(22)
	5%	DB	38.9	38.4	48.6	60.6	71.3	77.4	80.1	78.5	75.3	66.2	54.5	42.8	(23)
		MCWB	36.9	35.7	43.4	51.1	60.9	67.3	69.7	68.7	66.8	60.2	50.3	40.6	(24)
	10%	DB	35.8	35.4	43.1	55.1	67.0	74.0	77.3	75.9	71.9	61.7	50.4	39.4	(25)
		MCWB	34.0	33.2	38.7	47.0	58.0	65.0	67.8	67.3	65.0	56.7	46.8	37.0	(26)
	0.4%	WB	48.6	48.4	58.8	63.0	70.0	74.4	75.9	74.7	73.4	68.5	58.3	52.4	(27)
		MCDB	49.7	49.6	65.0	70.9	79.5	84.1	83.3	82.3	79.3	73.1	61.1	54.7	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.0	40.4	50.7	57.6	66.2	71.3	73.3	72.4	70.7	63.9	54.3	45.0	(29)
		MCDB	43.1	42.4	54.1	65.6	73.9	79.3	80.4	79.5	76.4	68.6	57.8	46.3	(30)
	5%	WB	37.2	36.0	44.1	52.3	62.8	68.8	71.3	70.6	68.2	60.4	50.6	40.9	(31)
		MCDB	38.7	37.8	48.0	59.3	69.5	75.5	78.0	76.4	73.2	65.3	54.1	42.5	(32)
	10%	WB	34.2	33.4	39.0	47.9	59.0	66.5	69.5	68.9	65.9	57.2	47.0	37.2	(33)
		MCDB	35.6	35.3	42.8	54.2	65.3	72.3	75.4	73.9	70.9	61.0	50.2	39.2	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	13.0	14.0	14.2	16.4	18.6	18.0	17.1	16.9	17.3	14.1	11.7	10.2	(35)
		MCDBR	13.9	15.2	20.4	27.1	24.9	22.0	20.4	20.3	20.8	20.2	18.1	13.3	(36)
		MCWBR	13.4	13.7	15.6	18.2	15.1	13.0	11.5	11.4	12.3	13.8	14.7	12.4	(37)
	5% WB	MCDBR	13.4	14.3	19.4	25.0	22.4	19.4	16.8	16.8	18.2	17.8	16.9	13.1	(38)
		MCWBR	13.3	13.4	16.1	18.4	15.5	12.6	10.7	10.7	12.2	13.3	14.8	12.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.285	0.300	0.306	0.351	0.372	0.401	0.404	0.399	0.367	0.350	0.316	0.283	(40)
	taud		2.416	2.330	2.418	2.353	2.381	2.318	2.349	2.348	2.426	2.440	2.485	2.474	(41)
	Ebn at Noon		266	281	296	289	285	276	273	270	271	260	252	256	(42)
	Edh at Noon		23	30	32	37	37	40	38	37	31	27	22	20	(43)
(44) All-Sky Solar Radiation	RadAvg		354	642	1079	1448	1752	1918	1938	1650	1273	742	407	256	(44)
	RadStd		45	73	116	131	163	134	93	82	115	82	69	31	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page